

Data Flow Diagram	
Date	15 March 2026
Team ID	SWTID-2026-9030
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	2 Marks

Data Flow Diagram

The DFD shows how information moves through LegalEase between the user, frontend, backend/API, AI model, document generator, and generated document outputs.

USER Document type Parties Terms Effective date	1. INPUT & VALIDATION Streamlit Frontend Collect and validate user inputs	2. DOCUMENT GENERATION FastAPI Backend POST /generate
	↓ validated request	↓ structured prompt
Document Request User-provided legal details	3. AI PROCESSING Gemini model Generate legal content	4. FORMATTING & PREVIEW Editable preview Terms table Branding
	↓ generated legal text	↓ formatted output
Generated Document Data Draft / clauses / terms	5. EXPORT TXT / DOCX / PDF	USER Edit and download

Supporting Components: Google Gemini API provides AI-based legal text generation. Document-formatting utilities create DOCX and PDF files, while TXT export provides plain text output.

Key Data Flows: User inputs → validation → Gemini prompt → generated legal content → editable preview → TXT/DOCX/PDF download.